A Dual Global–Local Attention Network for Hyperspectral Band Selection

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***Abstract*— This article proposes a dual global–local attention network (DGLAnet), which is an end-to-end unsupervised band selection (UBS) method that fully utilizes spatial and spectral information in both global and local aspects. The DGLAnet assumes that BS can be realized using the hyperspectral image (HSI) reconstruction process. First, the DGLAnet implements a dual attention module to obtain spatial–spectral and global–local features to reweight the HSI data. It adopts bi-directional relations to grasp spatial and spectral features from a global perspective. Meanwhile, the DGLAnet extracts local features through max-pooling and mean-pooling and then merges them via the convolution operation. Global–local features are utilized to learn attention to recalibrate the original data, and the reconstruction module is adopted to restore the original image from the reweighted HSI data. Finally, a proper band subset is selected by the constructed band evaluation index. Experiments on three hyperspectral data show that the DGLAnet outperforms other state-of-the-art methods and uses all bands with a lower computational cost.**

***Index Terms*— Band selection (BS), global–local attention, hyperspectral image (HSI), spatial–spectral features.**

I. Introduction

HYPERSPECTRAL image (HSI) contains abundant spectral information of ground objects through dozens or even hundreds of contiguous narrow bands [1]. With improved spectral resolution, HSI imaging can obtain rich spatial–spectral information simultaneously [2], [3], reflecting diagnostic features of ground objects [4]–[6]. Therefore, hyperspectral remote sensing is widely used in coastal wetland mapping [7], fine crop classification [8], planetary exploration [9], [10], water quality evaluation [11], geological exploration [12], and so on. However, HSI suffers from strong spectral redundancy [13]. In addition, an enormous amount of HSI bands results in a huge data volume, which will cause a great burden on data processing. High-dimensional characteristic also brings about "the curse of dimensionality," which may lead to the decline of the classification accuracy when training samples are limited [14], [15]. Therefore, it is necessary to reduce its dimension before application.

There are two-dimensionality reduction methods for HSI: feature extraction and band selection (BS). Feature extraction maps HSIs from high-dimensional space to low-dimensional space while retaining important information, such as sparse and low-rank near-isometric linear embedding (SLRNILE) [16]. Nevertheless, the feature extraction process will cause HSIs to lose their original physical meaning. Unlike feature extraction, BS is to select an appropriate band subset from the original ones. This approach retains the original band meanings and makes the result more interpretable [17]. Therefore, BS is focused on this research.

Depending on whether labeled samples are used, the BS can be divided into supervised, semisupervised, and unsupervised, which, respectively, indicate the use of labeled samples, the use both of labeled and unlabeled samples, and the use of unlabeled samples. Supervised methods select the band subset that can provide the highest classification accuracy by continuously training the classifier, for example, Spatialwrapper [18]. Semisupervised methods choose the appropriate bands by using the information of labeled and unlabeled samples at the same time, for example, dual-constrained low-rank representation (DCLRR) [19]. Although supervised and semisupervised methods can achieve excellent results, labeled samples may be difficult to collect. So, unsupervised methods are the hotspots of current studies. Unsupervised methods can be further divided into the following four types: ranking-, searching-, clustering-, and sparsity-based methods. In ranking-based methods, the HSI bands are evaluated and ranked according to a specific metric, for example, optimal band index based on fractal dimension [20] and enhanced fast density peak-based clustering (E-FDPC) [21], and the highly ranked bands are selected. The ranking-based methods independently estimate the importance of each band, which may result in a high similarity between the selected bands. The searching-based methods implement a search strategy to find appropriate bands and define the criterion function to achieve the optimal bands, such as linear prediction (LP) [22] and volume-gradient-based

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BS (VGBS) [23]. The computational cost of these methods is usually very high. Clustering-based methods divide the original bands into a number of clusters, and the bands are then selected from each cluster, for example, graph clustering [24] and optimal clustering framework (OCF) [25]. They effectively reduce the correlation between the selected bands, but cannot assure maximum information from the selected bands and are susceptible to noise. Sparsity-based methods regard the process of BS as an optimization problem with sparse constraints, for example, improved sparse subspace clustering (ISSC) [26] and fast and latent low-rank subspace clustering (FLLRSC) [27]. The quality of sparse coefficients generated by these methods depends on the optimization method, and the computational complexity is also high [28].

Recently, the emergence of deep learning led by convolutional neural network (CNN) has revolutionized the traditional study of HSI [29]. Deep learning originated from the artificial neural network and learns complex nonlinear features by stacking different levels of representation layers. These layers are not manual but learned from data. Deep learning often implements the backpropagation algorithm to instruct changes in internal parameters, thereby discovering complex structures within datasets [30], [31]. Since it has terrific nonlinear representation ability, it has been widely explored in HSI classification [32], [33], target detection [34], unmixing [35], and other aspects. Deep learning has also achieved some progress in BS [36]. For instance, BS based on CNN (BSCNN) selects the optimal bands using the classification accuracy of a well-trained CNN [37]. After that, end-to-end learning has been adopted to handle the time-consuming problem of BSCNN. For example, bandwise-independent convolution and hard thresholding (BHCNN) [38] embeds both BS and classification into a training network, and the ternary weight CNN (TWCNN) [39] was then presented to optimize the program. Moreover, to reduce the manual intervention, some scholars proposed unsupervised methods. Deep subspace clustering (DSC) [40] based on an autoencoder (AE) embeds subspace clustering into the training procedure to investigate proper bands with lower correlations. Concrete AE-based unsupervised band selection (CAE-UBS) [41] integrates the Gumbel-softmax operator with an AE to demonstrate the significance of each band. However, the black-box procedures of deep learning-based methods render the selected bands lack in interpretability.

In recent years, attention mechanism has become one of the research hotspots in deep learning. The attention mechanism allows deep learning networks to focus limited resources on more important parts and suppresses useless information such as the human visual system. This mechanism strengthens the interpretation of deep learning. Researchers have made some exploration work by integrating the attention mechanism into deep learning networks. For example, SpectralFormer [42], a transformer-based backbone network, is proposed for HSI classification. Fully contextual networks (FullyContNets) [43] use the scale attention module to adaptively acquire and aggregate features of different scales for HSI scene parsing.

The major objective of the attention mechanism is to select more critical information from a large amount of information. That fits well with BS which is to select the most representatives from numerous bands. A few investigators have combined the deep learning network with the attention mechanism for HSI BS. They can be roughly subdivided into two types, according to the network optimization method, that is, classification- and reconstruction-based. As for classification-based methods, the network is optimized by implementing the classification task. Wang *et al.* [44] adopted the band attention network to yield a fixed band subset. A dense spatial–spectral attention network (DSSAN) [45] that jointly uses spatial–spectral features is proposed for BS. The reconstruction-based methods treat BS as a reconstruction task and select the most representative bands to reconstruct the original data. BS network (BS-Net) [46] consists of a band attention module (BAM) and a reconstruction network (RecNet) for creating a flexible architecture. The nonlocal band attention network (NBAN) [47] brings the nonlocal attention mechanism into BS-Net to recalibrate the original HSI and select bands with larger weights. NBAN makes full use of long-range spectral information and overcomes its limitations in the local range of BS-Net. Similarly, the attention-based AE (AAE) [48] uses the AE globally to capture the nonlinear correlations between bands. Later, the dual-attention reconstruction network (DARecnet) [49] was presented to gain spatial–spectral self-attention features to benefit from selecting proper bands.

However, existing methods have the following two severe problems. One is that they only use either local or global features in the attention mechanism. In fact, both features are essential and complementary to each other. By global receptive field, global features provide more contextual information and can improve the network capacity. The local information is also critical for characterizing the property of the HSI [50]. Without a local-feature extraction module, fine-grained semantic information, for example, the similarity of adjacent bands and spatial correlations among nearby ground objects, is likely to be ignored [51], [52]. The lack of fine-grained semantic information would impact the correct allocation of attention weights and reconstruction of the HSI data, which in turn degrades the selected bands. The other is that these methods rarely capture and use spatial–spectral information together. Different ground objects have different discriminative spectral characteristics. Moreover, neighboring pixels usually have high probabilities of belonging to the same class of ground objects, and vice versa [53]. Lacking a spectral–spatial attention module, they cannot fully depict spatial relations of the main ground objects to find representative bands [49], [54].

Therefore, this article proposes a dual global–local attention network (DGLAnet) for HSI BS. It aims to grasp the overall and adjacent correlation of bands and to investigate the global–local discriminative characteristics of ground objects. The DGLAnet can effectively suppress irrelevant bands and choose the most representative band subset. In order to obtain global attention, spatial and spectral information is captured using bi-directional relations. It avoids the drawbacks of the convolution kernel to obtain information from the global receptive field that considers the intracorrelations among all the bands. Meanwhile, the local spatial–spectral features are obtained through the pooling layers. After that, both global and

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local features are stacked to obtain attention weights, which are optimized through a reweighted reconstruction process. Finally, the constructed band evaluation index is used to select bands. Our main contributions are as follows.

1) We are the first to introduce the idea of jointly extracting global–local deep inner information from both spatial and spectral aspects for HSI BS and propose a DGLAnet. DGLAnet can not only fully grasp the global discrimination information, but also take into account the local similarity of HSI. It extracts spatial features to represent the distribution characteristics of ground objects and their spatial relations.
2) DGLAnet integrates the spatial–spectral feature extraction module, the global–local information extraction module, and the HSI reconstruction module. It uses bi-directional relations [55] to obtain global spatial–spectral features. Max-pooling and mean-pooling are adopted to capture the local spatial–spectral features. Moreover, the evaluation index that combines the information entropy and the correlation coefficient is designed to select proper bands.
3) DGLAnet requires fewer parameters, which is more practical than other deep learning-based BS methods.

The remaining article is arranged as follows. Section II introduces the DGLAnet method. In Section III, several experiments on three datasets are carried out. Section IV concludes this article.

II. Proposed Method

Fig. 1 illustrates the proposed DGLAnet for HSI BS. It consists of a spatial–spectral attention module and a reconstruction network. After the DGLAnet is well trained, we use an evaluation index to assess and rank the bands and select the top-ranked ones as the finally selected bands.

Let a 3-D HSI data cube with height h, width w, and band b be denoted as $\mathbb{X} \in \mathbb{R}^{h\times w\times b}$. In the preprocessing, it is divided into several subcubes $\{X_i\}_{i=1}^{h\times w} \in \mathbb{R}^{s\times s\times b}$. From the spatial perspective, each subcube has $s \times s$ b-dimensional features $\{x_m \in \mathbb{R}^b\}_{m=1}^{s\times s}$. From the spectral perspective, each subcube has b s^2-dimensional features $\{x_p \in \mathbb{R}^{s\times s}\}_{p=1}^{b}$, in which $s \times s$ is the size of the neighboring window.

A. Spatial Global–Local Attention

Spatial global–local attention (Spatial-GLA) [55] learns the spatial attention of HSI by suppressing the unimportant spatial parts and enhancing the significant parts. It takes the subcube $X_i = \{x_m \in \mathbb{R}^b\}_{m=1}^{s\times s}$ as the input of the attention module and outputs the Spatial-GLA.

As shown in Fig. 1, Spatial-GLA includes two components, which learn the global and local spatial features, respectively. After the subcube is fed into Spatial-GLA, the function $G_s(\cdot)$ is used to measure the affinity between spatial features as

$$\mathrm{SR}_{m,n} = G_s(x_m, x_n) = \Theta_s\left(x_m; \theta_{\Theta_s}\right)^{\mathrm{T}} \Phi_s\left(x_n; \theta_{\Phi_s}\right) \tag{1}$$

where $\Theta_s(\cdot)$, $\Phi_s(\cdot)$ are the embedding functions and θ_{Θ_s} and θ_{Φ_s} are the parameters to be trained in Θ_s and Φ_s, respectively. The output of $\Theta_s(\cdot)$ and $\Phi_s(\cdot)$ are FM1 and FM2, respectively. The structure of Θ_s, Φ_s includes a 1×1 convolutional layer, a max-pooling layer followed by a batch normalization (BN) layer, and a parametric rectified linear unit (PReLU) layer. The affinity matrix $\mathrm{AM}_s \in \mathbb{R}^{s\times s}$ is constructed using the affinity relations of spatial features, that is, FM1 and FM2. Using the affinity matrix, we can obtain the bi-directional affinity relations of each spatial feature, that is, $\mathrm{RV}_m = [\mathrm{AM}_s(m,:), \mathrm{AM}_s(:,m)]$. The set of bi-directional affinity relations $\mathrm{RV}^s = \{\mathrm{RV}_m\}_{m=1}^{s\times s}$ is the global spatial feature.

Local information can provide significant information and complement global information. In this article, we jointly consider the global and local information about HSI. While learning the global relation feature, the subcube is also fed into the local feature extraction module

$$\begin{aligned} L_s &= E_s\left(X_i; \theta_{L_s}\right) \\ &= \mathrm{Conv}_2(\mathrm{CAT}(\max(S), \mathrm{mean}(S))) \end{aligned} \tag{2}$$

$$S = \mathrm{Conv}_1(X_i) \tag{3}$$

where $E_s(\cdot)$ is the embedding function and θ_{L_s} are the parameters to be trained. After the subcube is sent to $E_s(\cdot)$, it first passes through a 1×1 convolutional layer $\mathrm{Conv}_1(\cdot)$. Then, we apply max-pooling and mean-pooling layers to extract texture features and background features, respectively. It has been confirmed that it is better to use these features at the same time [56]. After that, we concatenate texture features and background features. For the intention of producing the spatial local feature, a 1×1 convolutional layer $\mathrm{Conv}_2(\cdot)$ is used to further aggregate the concatenated features and excavate deeper information. The output of the 1×1 convolution layer is the spatial local feature. The BN layer and the PReLU activation layer are at the end of the convolutional layers.

Spatial-GLA stacks the global relation feature RV^s and the local feature L_s to obtain the spatial feature of SF

$$\mathrm{SF} = \mathrm{CAT}\left[(L_s), U_s\left(\mathrm{RV}^s\right)\right]. \tag{4}$$

Since the two feature dimensions are not the same, the dimension is unified through the embedding function $U_s(\cdot)$ before stacking. In order to obtain Spatial-GLA, the stacked spatial feature SF is sent to the attention module

$$A_S = \mathrm{Sigmoid}(\mathrm{sa}(\mathrm{SF}; \theta_{\mathrm{sa}})) \tag{5}$$

where $\mathrm{Sigmoid}(\cdot)$ is the activation function, which is used to map the input data to the range of 0–1 to ensure the nonnegativity of the attention weight, $\mathrm{sa}(\cdot)$ is used to learn the attention weight of the spatial part, and θ_{sa} is the parameter to be trained. $\mathrm{sa}(\cdot)$ contains two 1×1 convolutional layers. After passing through the convolutional layers, the band dimension of the spatial relation attention is reduced to 1.

B. Spectral Global–Local Attention

We designed spectral global–local attention (Spectral-GLA) [55], which is used to obtain the attention of the bands, and its structure is similar to Spatial-GLA. In Spectral-GLA, we regard the subcubes as having b s^2-dimensional features, that is, $X_i = \{x_p \in \mathbb{R}^{s\times s}\}_{p=1}^{b}$.

As demonstrated in Fig. 1, Spectral-GLA has two components, which can learn global and local spectral features.

Similar to the spatial global feature extraction module, the subcube is passed through an embedding function $G_b(\cdot)$

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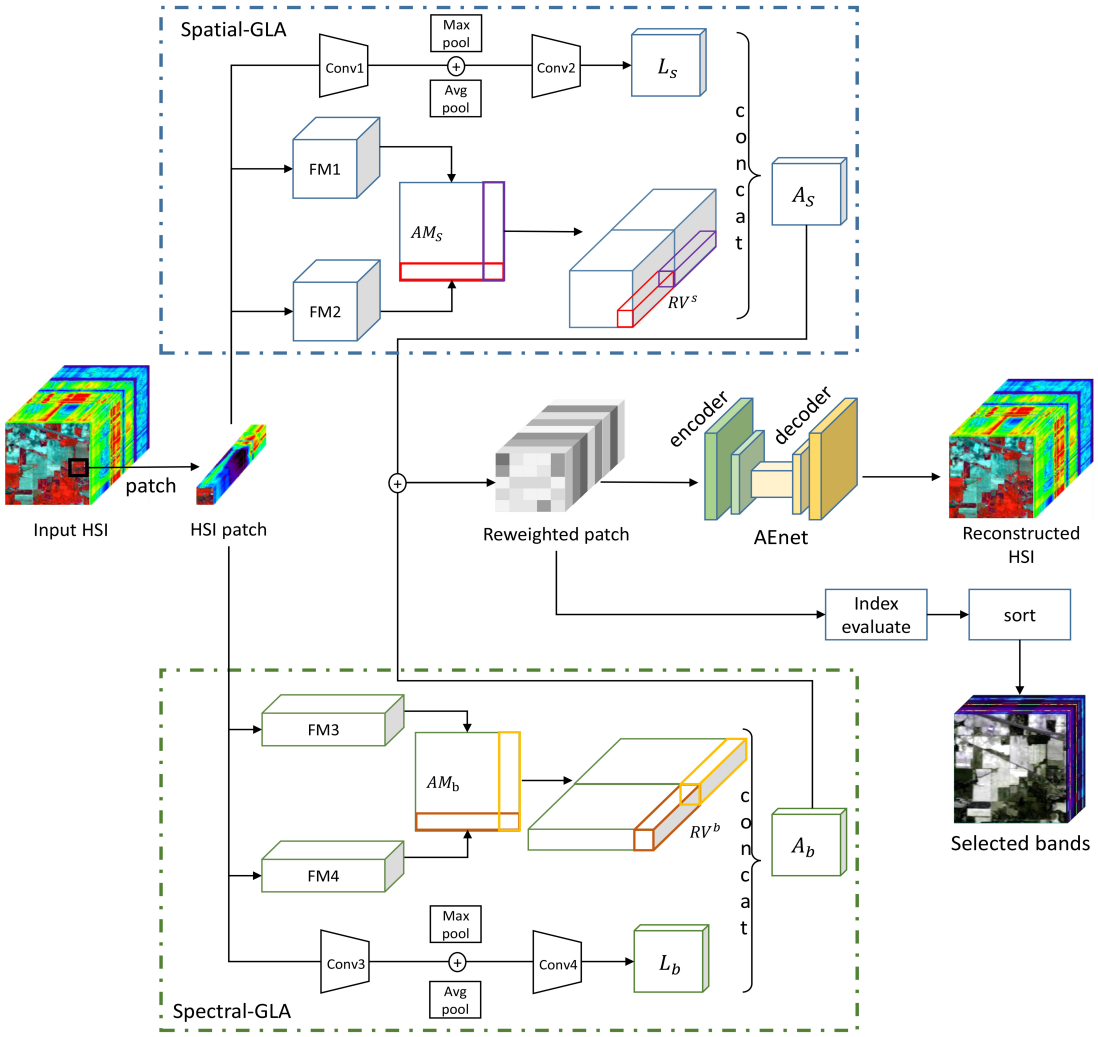


Fig. 1. Flowchart of DGLAnet for BS. We feed subcube into the embedding function to get the feature maps, that is, FM1–4. After that, the affinity matrices, that is, AM-s and AM-b, can be constructed using feature maps. And then global features are obtained from affinity matrices. Meanwhile, 1×1 convolutional layers and pooling layers are used to attain local features.

to generate the affinity between bands to represent global information

$$\mathrm{BR}_{p,q} = G_b(x_p, x_q) = \Theta_b(x_p; \theta_{\Theta_b})^T \Phi_b(x_q; \theta_{\Phi_b}) \quad (6)$$

where $\Theta_b(\cdot)$, $\Phi_b(\cdot)$ are the embedding functions and $\theta_{\Theta_b}, \theta_{\Phi_b}$ are the parameters to be trained in Θ_b, Φ_b. The structure of Θ_b, Φ_b includes a 1×1 convolutional layer followed by a BN layer and a PReLU activation layer. FM3 and FM4 are the output of $\Theta_b(\cdot)$ and $\Phi_b(\cdot)$, respectively. Then the affinity matrix, that is, $\mathrm{AM}_b \in \mathbb{R}^{b \times b}$, can be obtained by FM3 and FM4. From the affinity matrix, we can obtain the bi-directional affinity relations of each band, that is, $\mathrm{RV}_p = [\mathrm{AM}_b(p, :), \mathrm{AM}_b(:, p)]$. The set of bi-directional affinity relations $\mathrm{RV}^b = \{\mathrm{RV}_p\}_{p=1}^{b \times b}$ is the global relation spectral feature.

The subcube is also used to capture local spectral information by implementing the local spectral feature extraction module

$$\begin{aligned} L_b &= E_b(X_i; \theta_{L_b}) \\ &= \mathrm{Conv}_4(\mathrm{CAT}(\max(B), \mathrm{mean}(B))) \quad (7) \\ B &= \mathrm{Conv}_3(X_i) \quad (8) \end{aligned}$$

where $E_b(\cdot)$ is the embedding function and θ_{L_b} are the parameters to be trained. Like Spatial-GLA, after the subcube is fed into $E_b(\cdot)$, it first passes through a 1×1 convolutional layer $\mathrm{Conv}_3(\cdot)$. Then the salient features and the background features between the bands are extracted by the max-pooling layer and mean-pooling layer. After that, $\mathrm{Conv}_4(\cdot)$ is used to aggregate the features and generate the spectral feature. The BN layer and the PReLU activation layer are followed at the end of the convolutional layers.

Similar to (4) and (5), the obtained global and local information of bands is stacked to learn the spectral global–local attention map.

C. Overall Architecture

In this article, we treat BS as an HSI reconstruction task from critical bands. As shown in Fig. 1, DGLAnet has three main steps. First, DGLAnet applies spatial and spectral global–local attention modules to learn attention maps to adjust the original HSI. The purpose of the attention module is to identify representative bands. Second, we send the reweighted HSI into the reconstruction network. The reconstruction network

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is used to rebuild the origin HSI and construct the loss function. Finally, with the well-optimized DGLAnet, we use an evaluation index to rank the bands and select the top-ranked.

We use the attention mechanism to explore the decisive information of HSI. The essence of the attention mechanism is assigning a series of weights to the input for emphasizing the important parts and enabling the model to be more accurate. The spatial and spectral attention weights of HSI are learned through the attention module mentioned above and merged to obtain the dual global–local attention (DGLA)

$$A_{\mathrm{DGLA}} = A_s(X_i;\theta_s) \oplus A_b(X_i;\theta_b) \tag{9}$$

where A_s, A_b are global–local attention learning modules, θ_s, θ_b are the parameters to be trained, and $\oplus$ is the elementwise addition operation. By doing so, we can find spatial–spectral information from global and local perspectives.

Then we use the obtained DGLA map to reweight the HSI. In this way, the important information can be highlighted. Following the HSI reweighting operation, we use the learned important bands to restore the original HSI:

$$\mathcal{X}_i = A_{\mathrm{DGLA}} \otimes X_i \tag{10}$$

$$\widehat{\mathcal{X}_i} = F_{\mathrm{rc}}(X_i;\theta_{\mathrm{rc}}) \tag{11}$$

where $\otimes$ is the Hadamard product, $F_{\mathrm{rc}}(\cdot)$ is the reconstruction network, and θ_{rc} are the parameters to be trained. The reconstructed network is composed of two convolutional layers, two deconvolutional layers, and a max-pooling layer that connects the layers mentioned above. All the kernel sizes of convolutional and deconvolutional are $1 \times 3 \times 3$. Each convolutional layer and deconvolutional layer is followed by BN and PReLU activation layers. We use the loss function to evaluate the effect of network reconstruction

$$\mathrm{Loss} = \frac{1}{2\mathrm{ts}} \sum_{i=1}^{\mathrm{ts}} \|\mathbb{X}_i - \widehat{\mathcal{X}}\|_1 \tag{12}$$

where ts is the number of samples. The optimization of the network is completed by minimizing the loss function. We employ stochastic gradient descent (SGD) with moment [57] to improve the convergence (12).

In order to select the most representative band, we present the band evaluation index (IC). We, respectively, calculate and normalize the information entropy and correlation coefficient of the bands $\{b_i\}_{i=1}^{b}$ and define their ratio as the band evaluation index (IC), which are defined as

$$\mathrm{IC} = \frac{\mathrm{IE}(b_i)}{\mathrm{CC}(b_i)} \tag{13}$$

$$\mathrm{IE}(b_i) = \frac{I(b_i)}{\sum_{i=1}^{b} I(b_i)} \tag{14}$$

$$\mathrm{CC}(b_i) = \frac{C(b_i)}{\sum_{i=1}^{b} C(b_i)} \tag{15}$$

where

$$I(b_i) = -\sum_{g} p(g)\log(p(g)) \tag{16}$$

$$C(b_i) = \sum_{j=1}^{b} \frac{\sum_{l=1}^{s\times s}\left(b_{il}-\bar{b}_i\right)\left(b_{jl}-\bar{b}_j\right)}{\sqrt{\sum_{l=1}^{s\times s}\left(b_{il}-\bar{b}_i\right)^2 \sum_{l=1}^{s\times s}\left(b_{jl}-\bar{b}_j\right)^2}} \tag{17}$$

Algorithm 1: Pseudocode of DGLAnet

1: Input: HSI data $\mathbb{X} \in \mathbb{R}^{h\times w\times b}$, patch size of HSI cube s, hyper-parameters of SGD, the number of selected bands k
2: Preprocess the Original HSI data $\mathbb{X} \in \mathbb{R}^{h\times w\times b}$ to generate the HSI cubes $X_i \in \mathbb{R}^{s\times s\times b}$
3: Random initialize the to be trained parameters θ_s, θ_b *and* θ_{rc}
While *maximum iteration is met* **do**
  1) extract the spatial attention A_S and spectral attention A_b
  2) fuse the spatial attention A_S and spectral attention A_b to generation the DGLA $A_{\mathrm{DGLA}} = A_s(X_i;\theta_s) \oplus A_b(X_i;\theta_b)$
  3) reweight the original HSI data $\chi_i = A_{DGLA} \otimes X_i$
  4) reconstruct the original HSI data $\widehat{\mathcal{X}_i} = F_{rc}(\chi_i;\theta_{rc})$
  5) calculate the loss and optimize it using SGD
End
4: Calculate IC value of each band of reweighted HSI
5: Select top k bands with larger IC values
Output: the band subset

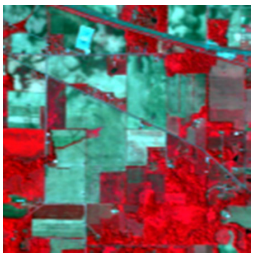


Fig. 2. Indian Pines dataset.

where g denotes a gray level of the histogram in b_i consisting of $s \times s$ pixels, $p(g)$ is the probability of g in the histogram of b_i, b_{il} and b_{jl} are the lth pixel values of b_i and b_j, and $\bar{b}_i$ and $\bar{b}_j$ are the average values of all pixels in the ith and jth bands, respectively. A larger IC means the corresponding band is more important.

The pseudocode of DGLA-net is summarized in Algorithm 1.

III. Experimental Results

A. Datasets

The Indian Pines dataset was acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site in North-western Indiana, USA. It has 145145 pixels and its spatial resolution is 20 m. It consists of 224 spectral bands ranging from 400 to 2500 nm. Due to the influence of water absorption and noise, 24 bad bands (104–108, 150–163, 220) were removed and 200 bands remained for processing. The dataset has 16 types of ground objects and its false-color image is shown in Fig. 2.

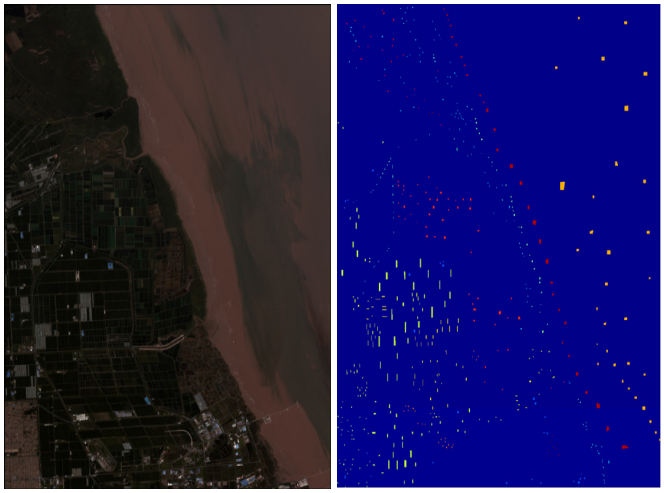
The PaviaU dataset in Fig. 3 was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor during a flight campaign over Pavia, northern Italy. It consists of 610610 pixels, which becomes 610340 after removing the samples without information. The spatial resolution is 1.3 m. It has 103 spectral response bands in the spectrum range of 430–860 nm.

The Yancheng dataset is an image of the coastal area of Yancheng, Jiangsu, China, obtained by the ZY1_02D_AHSI sensor on September 6, 2020. The image size is 1398×942, which is much larger than the Indian Pines and the PaviaU. Its spatial resolution is 30 m. The wavelength of this dataset ranges from 395 to 2500 nm. After removing the 19 bad bands, that is, 22–27, 49–57, 87–90, the Yancheng data

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Fig. 3. PaviaU dataset.



(a) (b)

Fig. 4. Yancheng dataset. (a) False-color composite image. (b) Ground truth.

retains 147 bands. This area is rich in wetland resources with 16 classes of ground objects. Fig. 4 presents the composite image and ground truth of the Yancheng dataset.

B. Experimental Setup

We compare the presented method with seven state-of-the-art methods and using all bands, including four traditional methods, for example, LP (search-based) [22], ISSC (Sparsity-based) [26], E-FDPC (ranking-based) [21], OCF (clustering-based) [25], and three deep learning-based methods, for example, BS-Conv [58], BS-fully connected network (FC) [46], and DARecnet [49].

The hyperparameters in the method proposed in this article include learning rate (*lr*) and batch size. Referring to the method in [59], we determine that the learning rates of Indian Pines, PaviaU, and Yancheng data as 0.05, 0.04, and 0.02, respectively. Combining the suggestions in [60] and the performance of the hardware, we set the batch size as 64 and the number of iterations as 100. The parameter settings in the comparison methods follow its default settings. The iterations of finding the best starting band in LP are set to be 100. We set the scalar parameter in ISSC to 0.005. In OCF, the top-rank cut (TRC) and the E-FDPC are adopted as the objective function and ranking method, respectively. The learning rate, regularization coefficient, and iterations of BS-Conv and BS-FC are set to be $1e^{-2}$, $2e^{-3}$, and 100, respectively.

We use the support vector machine (SVM) [61] as the classifier, which adopts the radial basis function (RBF) as a kernel. The variance parameter and penalization factor used in SVM are determined via threefold cross-validation. Overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa) are used to evaluate the classification accuracies. We randomly select 10% of the labeled samples in the Indian Pines dataset as the training samples. For the PaviaU data, we set the training sample ratio to be 5%. For the Yancheng data, we choose a 1% training sample ratio and the rest for testing. The neighboring window size is 5×5. All experiments are independently repeated ten times, and the averages are used as the final result.

DGLAnet and the two methods based on deep learning are implemented in Python 3.6 using Intel Core i7-10700 2.90 GHz CPU with 128-GB RAM and NVIDIA GeForce RTX 3080.

C. Classification Performance of Different Methods

This experiment analyzes the effectiveness of different methods by changing the number of selected bands from 5 to 50 with a step interval of 5.

Fig. 5 shows the OA of the eight methods and all bands on the three datasets. Because the result of AA and kappa is similar to OA, we did not exhibit the curves of AA and kappa. The averaged AA, OA, and kappa of different methods are listed in Table I. For the Indian Pines dataset in Fig. 5, when the number of selected bands is greater than 35, the curve of DGLAnet is better than other methods and all bands. With a small number of selected bands, for example, ten, DGLAnet can also achieve good classification accuracy, closing to all bands. The other three deep learning-based methods are excellent as well, and their classification accuracy exceeds the one from using all bands. Among them, the curve of DARecnet is slightly higher than that of BS-Conv and BS-FC. The performance of ISSC is the second only to deep learning-based methods and is the best and most stable among traditional methods. When the number of selected bands increases, the accuracy of LP continues to rise, but the curse of dimensionality occurs on the OCF.

As regards the PaviaU dataset, combined with the results in Fig. 5 and Table I, DGLAnet has the best classification behavior and is more stable than other methods. When the number of bands is greater than 40, the performance of DGLAnet is close to or even better than those of all bands. The curves of other methods are close to that of DGLAnet, and the mean accuracies of other methods also produce satisfactory results. On the whole, ISSC achieves the second-best performance but its curve is slightly lower than LP. BS-FC attains noticeable precision when the number of selected bands is greater than 40. The performance of BS-Conv and DARecnet is not outstanding and unstable, although there are occasional highlights.

As for the Yancheng dataset, although OCF performs the best when the number of selected bands is 10, 20, and 40, its performance is not stable. The DGLAnet is always at the forefront, and the OA curve is significantly higher than other methods. That indicates the proposed DGLAnet is more robust. BS-FC, which is also based on deep learning, performs quite well. BS-FC has a higher curve than the majority of the contrast methods. The other two deep learning-based methods perform poorly. This may be because DGLAnet can jointly

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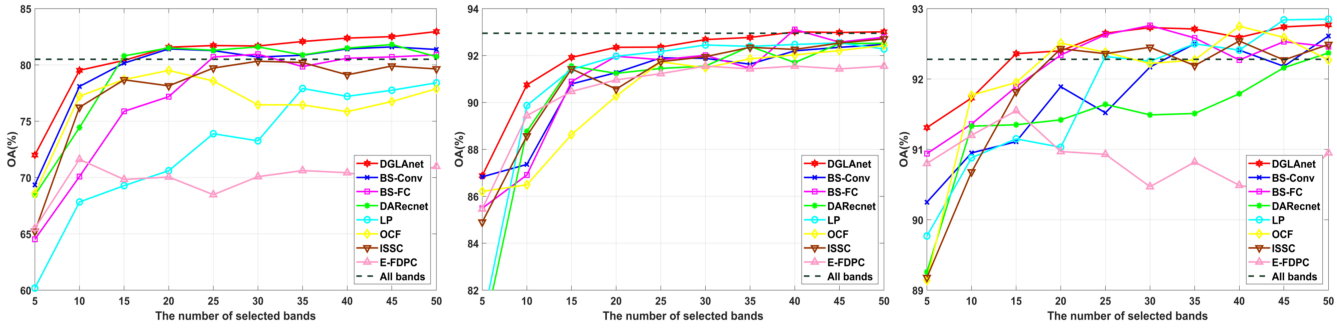


Fig. 5. Classification accuracy of different methods under different number of bands. (a) Indian Pines. (b) PaviaU. (c) Yancheng.

TABLE I

CLASSIFICATION ACCURACY OF DIFFERENT METHODS UNDER DIFFERENT NUMBER OF BANDS (AVERAGE VALUE OF 5–50)

Datasets	Indices	E-FDPC	LP	ISSC	OCF	BS-FC	BS-Conv	DARecnet	DGLAnet
Indian Pines	AA	61.6±3.44	65.92±5.81	72.29±5.83	70.81±4.24	72.54±5.72	73.38±4.87	74.85±5.74	**76.91±2.81**
	OA	69.74±1.62	72.63±5.51	77.72±4.32	76.6±2.89	77.85±5.09	79.1±3.47	79.3±4.17	**80.68±3.05**
	kappa	0.652±0.019	0.686±0.064	0.745±0.051	0.732±0.034	0.75±0.059	0.759±0.04	0.763±0.048	**0.779±0.035**
PaviaU	AA	87.85±1.83	88.09±4.19	88.16±3.06	87.65±2.43	88.47±2.75	88.16±2.16	87.28±4.95	**89.27±2.34**
	OA	90.51±1.8	90.81±3.5	90.9±2.31	90.33±2.26	90.90±2.58	90.87±1.95	90.36±3.63	**91.87±1.79**
	kappa	0.873±0.025	0.877±0.049	0.879±0.032	0.871±0.031	0.88±0.036	0.878±0.027	0.871±0.051	**0.892±0.02**
Yancheng	AA	75.82±1.32	79.11±1.65	79.66±1.36	79.85±1.08	79.45±1.61	79.62±1.17	79.20±1.73	**80.58±1.11**
	OA	90.86±0.34	91.8±0.97	91.84±1.03	91.99±0.99	92.19±0.61	91.76±0.74	91.43±0.79	**92.39±0.47**
	kappa	0.893±0.004	0.904±0.011	0.904±0.012	0.906±0.011	0.91±0.008	0.903±0.008	0.899±0.009	**0.911±0.006**

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT GROUND OBJECTS IN THE INDIAN PINES DATASET

No	E-FDPC	LP	ISSC	OCF	BS-FC	BS-Conv	DARecnet	DGLAnet	All bands
1	34.96±10.97	35.37±7.01	52.03±12.96	70.73±5.97	63.9±19.29	53.66±5.59	**87.8±4.88**	87.8±5.01	60.98±13.36
2	59.17±5.8	60.95±2.84	75.28±1.8	76.52±0.46	77.31±2.09	79.69±1.07	76.54±1.98	74.4±0.75	**80.08±2.04**
3	55.96±4.03	55.46±4.56	64.44±3.88	56.72±4.39	63.61±3.53	61.98±4.93	63.56±2.87	61.98±4.59	**68.54±3.55**
4	35.84±10.21	45.77±3.33	54.93±4.52	**69.48±4.32**	60.94±2.17	53.05±4.92	62.25±5.91	66.2±4.79	56.34±5.05
5	77.7±3.34	87.01±2.45	83.52±2.01	83.07±3.69	86.85±1.68	**94.02±1.96**	90.99±2.73	86.21±1.47	86.21±1.47
6	88.48±2.06	94.37±0.54	96.25±1.09	92.24±1.63	92.97±3.58	89.35±1.85	96.47±0.73	**96.65±1.11**	96.65±1.39
7	30.67±8.22	49±6.56	70.67±18.86	58.67±19.14	63.2±14.4	**88.00±5.92**	71.2±10.68	88.00±6.53	68.00±21.92
8	95.04±2.94	95.12±1.58	98.29±1.16	97.21±0.95	96.84±1.03	**98.60±1.01**	96.84±2.02	96.98±1.16	97.21±1.52
9	35.19±11.42	25±12.11	27.78±9.07	37.04±22.38	41.11±15.15	**83.33±11.79**	52.22±18.87	50±13.61	33.33±15.20
10	45.94±6.73	68±3.58	76.61±2.38	69.26±1.63	77.21±2.46	77.03±1.57	74.63±1.98	**77.83±1.75**	75.43±2.21
11	72.2±4.03	77.35±1.72	80.01±0.67	77.59±0.74	81.2±1.53	82.07±0.88	81.48±1.48	**82.8±0.96**	79.90±1.61
12	56.74±5.73	53.79±6.53	75.97±3.29	72.47±4.38	74.04±2.2	73.41±1.78	80.6±4.09	**84.27±1.08**	61.24±4.54
13	96.56±1.36	94.16±3.18	95.47±1.02	91.49±2.23	96.52±0.43	94.57±0.71	96.41±3.16	**99.46±0.77**	95.65±1.39
14	91.97±4.03	94.84±1.2	94.17±0.77	93.67±0.76	95.1±1.43	95.87±0.82	93.87±1.26	**96.49±0.9**	95.96±1.19
15	34.77±3.59	47.69±3.78	48.41±1.84	**59.65±1.93**	57.23±3.12	56.77±3.58	59.08±4.56	50.14±2.12	57.93±4.74
16	70.24±5.14	83.63±3.89	77.78±8.49	84.52±4.45	78.57±5.16	**89.29±9.01**	75±2.48	79.76±9.03	83.33±4.64
AA	61.34±1.83	66.75±1.26	73.23±0.84	74.75±2.35	75.41±2.84	79.42±0.93	78.68±1.44	**79.9±0.98**	74.8±2.31
OA	68.48±0.71	73.89±0.88	79.72±0.82	78.57±0.64	80.7±0.56	81.27±0.39	81.3±0.37	**81.72±0.43**	80.5±0.51
Kappa	0.638±0.008	0.701±0.001	0.768±0.009	0.755±0.007	0.78±0.006	0.786±0.44	0.786±0.004	**0.791±0.004**	0.778±0.005

extract spatial and spectral features of the global–local. Due to the small proportion of selected samples, the accuracy of E-FDPC decreases when the number of selected bands is larger than 15.

According to [46], [49] and the classification performance of Fig. 5, unless otherwise specified, we select 25 bands in Indian Pines, 15 bands in PaviaU, and 30 bands in Yancheng for follow-up experiments. We demonstrate the classification accuracy of different ground objects on the three datasets in Tables II–IV, where bold fonts indicate the best results.

For the Indian Pines dataset, the four deep learning-based methods have distinct advantages over the traditional methods, and they are superior to using all bands. DGLAnet has the highest AA, OA, and kappa. Compared with all bands, DGLAnet increases by 5.1%, 1.2%, and 0.013 on AA, OA, and kappa, respectively. DGLAnet achieves the best classification

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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT GROUND OBJECTS IN THE PAVIAU DATASET

No	E-FDPC	LP	ISSC	OCF	BS-FC	BS-Conv	DARecnet	DGLAnet	All bands
1	88.54±0.61	90±1.06	88.54±0.61	**90.52±0.47**	89.05±1.53	89.96±1.02	90.5±0.97	89.57±0.71	89.81±0.74
2	97.21±0.7	96.99±0.38	97.21±0.7	96.11±0.48	96.52±0.56	97.3±0.4	96.65±0.45	97.32±0.24	**97.54±0.29**
3	76.38±5.11	74.43±2.73	76.38±5.11	65.26±4.96	70.91±0.32	77.53±5.2	**79.2±3.23**	70.79±5.03	69.81±1.73
4	90.76±1.45	91.38±1.51	90.76±1.45	90.42±1.53	91.41±1.06	88.81±1.82	85.77±1.8	90.45±0.91	**95.95±0.88**
5	98.54±1.21	98.61±0.13	98.54±1.21	98.38±0.9	98.49±0.1	98.15±1.48	98.57±0.31	**98.72±0.54**	98.28±0.28
6	83.67±2.48	82.88±1.2	83.67±2.48	66.09±3.63	80.95±0.74	78.89±1.8	85.73±1.62	**88.02±0.58**	86.25±1.09
7	82.12±2.35	80.25±2.77	82.12±2.35	79.36±2.01	81.89±5.27	75.22±3.93	79.1±2.12	**83.36±1.91**	74.27±1.56
8	85.74±2.99	86.36±0.84	85.74±2.99	86.94±1.43	88.45±1.29	85.39±3.01	85.6±2.5	86.01±2.08	**91.91±0.99**
9	99.73±0.18	99.62±0.11	99.73±0.18	99.48±0.34	99.48±0.19	99.62±0.18	99.83±0.15	99.73±0.18	**99.78±0.15**
AA	89.19±0.34	88.95±0.13	84±0.67	85.84±0.55	88.57±0.27	87.88±0.27	89.06±0.35	**89.33±0.5**	89.28±0.28
OA	91.43±0.21	91.41±0.28	88.16±0.23	88.63±0.4	90.89±0.14	90.79±0.17	91.49±0.13	91.92±0.16	**92.40±0.18**
Kappa	0.886±0.003	0.886±0.004	0.841±0.003	0.847±0.006	0.879±0.002	0.877±0.002	0.887±0.001	0.893±0.002	**0.899±0.002**

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT GROUND OBJECTS IN THE YANCHENG DATASET

Classes	E-FDPC	LP	ISSC	OCF	BS-FC	BS-Conv	DARecnet	DGLAnet	All bands
1	90.27±7.19	90.27±9.52	85.8±19.2	**95.02±2.52**	90.7±9.55	92.71±7.44	75.64±14.2	92.7±7.14	90.87±11.01
2	43.94±15.9	**64.36±15.59**	44.44±13.82	57.87±20.23	57.85±18.1	46.01±15.2	52.13±11.81	63.83±14.15	50.85±19.09
3	87.52±6.5	93.05±5.28	92.98±6.39	92.41±7.47	87.78±4.44	93.05±5.89	**98.21±1.17**	95.81±3.92	90.56±4.93
4	78.16±21.95	92.09±4.46	**93.97±3.62**	93.68±5.74	87.13±21.29	91.67±3.81	80.78±6.32	81.73±15.22	92.24±13.77
5	65.33±18.41	75.75±13.85	75.68±24.75	75.85±18.85	**82.84±6.61**	75.77±14	82.19±5.07	64.96±25.45	79.72±8.27
6	44.57±33.16	55.58±40.83	63.35±26.18	50.95±33.38	69.55±23.1	61.39±33.11	62.8±31.77	**81.36±18.05**	52.01±35.19
7	77.47±35.69	94.33±15.03	98±3.53	**99.18±1.22**	93.72±9.9	96.35±8.53	96.51±5.17	95.28±10.05	97.42±5.32
8	98.31±2.73	99.42±1.35	99.88±0.25	99.93±0.17	99.99±0.02	99.39±1.61	**100±0**	99.76±0.58	99.41±1.15
9	93.78±2.52	95.61±2.51	94.7±2.52	93.47±1.06	93.88±4.37	93.43±4.01	93.33±2.53	95.65±2.71	**96.55±3**
10	69.71±9.41	74.65±10.34	71.64±8.28	72.4±8.03	68.87±9.29	69.51±7.68	74.07±3.3	72.24±8.55	**76.7±8.38**
11	**100±0**	99.92±0.25	99.66±0.78	**100±0**	**100±0**	99.99±0.02	99.71±0.73	**100±0**	99.85±0.45
12	31.94±9.67	37.22±15.58	39.12±10.39	42.5±15.07	**52.95±11.55**	44.79±9.85	47.31±15.49	44.54±9.88	38.54±20.17
13	**96.13±1.94**	92.21±6.77	92.76±4.25	92.1±4.18	94.45±4.96	90.61±6.89	85.94±7.65	89.61±6.63	95.02±4.1
14	94.35±3.9	90±19.47	97.49±2.37	96.35±5.98	94.46±3.39	97.88±2.82	**98.86±1**	**98.88±0.8**	92.7±7.1
15	98.48±2.61	97.86±3.55	98.65±1.46	95.82±5.68	**98.65±1.31**	98.17±1.57	96.39±3.9	96.94±1.96	94.83±5.7
16	15.07±15.1	14.78±15.65	31.51±22.15	32.09±16.82	24.44±20.95	**37.69±13.55**	27.61±16.09	27.29±18.82	24.63±18.09
AA	74.07±2.85	79.19±4.28	79.98±2.65	80.6±3.05	81.08±3.7	80.53±2.45	79.47±3.43	**81.29±2.33**	79.49±1.87
OA	90.47±1.67	92.25±2.13	92.45±1.17	92.22±0.81	**92.76±0.98**	92.17±1.41	91.49±1.39	92.72±0.73	92.28±0.92
Kappa	0.888±0.019	0.909±0.025	0.911±0.014	0.909±0.01	0.915±0.012	0.908±0.017	0.9±0.016	**0.915±0.009**	0.909±0.011

accuracy in six types of ground objects, and most of the rest are also at the forefront.

From Table III, except AA of DGLAnet is slightly higher than that of all bands, eight comparison methods are inferior to all bands. This may be due to the superb image quality of PaviaU, with a spatial resolution of 1.3 m and little band noise. Among the comparison methods, the DGLAnet offers the best results and wins in three classes. DARecnet, which is also a deep learning-based method, comes in second. However, the results of BS-Conv and BS-FC are not ideal. E-FDPC and LP achieve good results as DARecnet. In terms of the Yancheng dataset, DGLAnet also performs admirably, with the best results of the three classes. BS-FC achieves the best in four categories and OA, but its AA and kappa are inferior to those of DGLAnet. Regarding the other deep learning-based methods, BS-Conv performs much better than DARecnet. Among traditional methods, the ISSC is similar to DGLAnet, and it outperforms the case of using all bands.

Furthermore, we compare the classification accuracies of eight methods with different training samples (Fig. 6). The ratio of training samples is changed in the range of 5–50 with a step interval of 5. As shown in Fig. 6, in the Indian Pines dataset, DGLAnet is superior to other methods. Similar to the state of Fig. 5, followed by OCF and ISSC, the four deep learning-based methods behave significantly better than the other four traditional methods. These methods are also prominent in the PaviaU dataset. Although OCF has an excellent result in the Indian Pines dataset, it falls short of the other rivals in the PaviaU dataset. ISSC also fails on the PaviaU dataset. LP achieves similar performance to the

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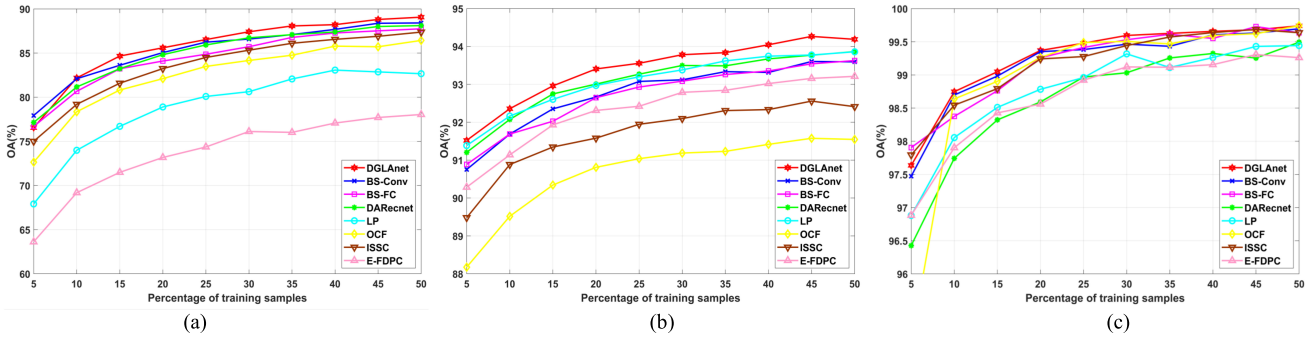


Fig. 6. Classification accuracy of different methods under different training samples. (a) Indian Pines. (b) PaviaU. (c) Yancheng.

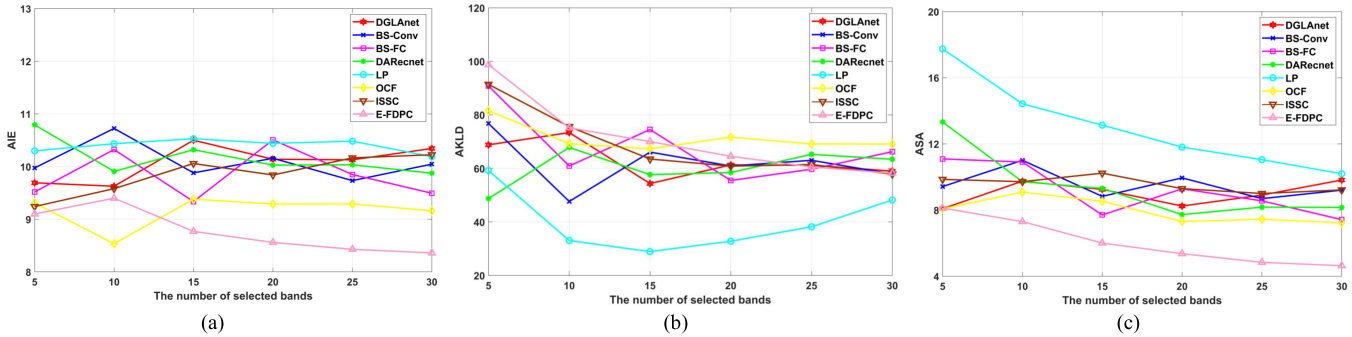


Fig. 7. Comparison in metrics of selected band subsets by different methods on the Indian Pines dataset. (a) AIE. (b) AKLD. (c) ASA.

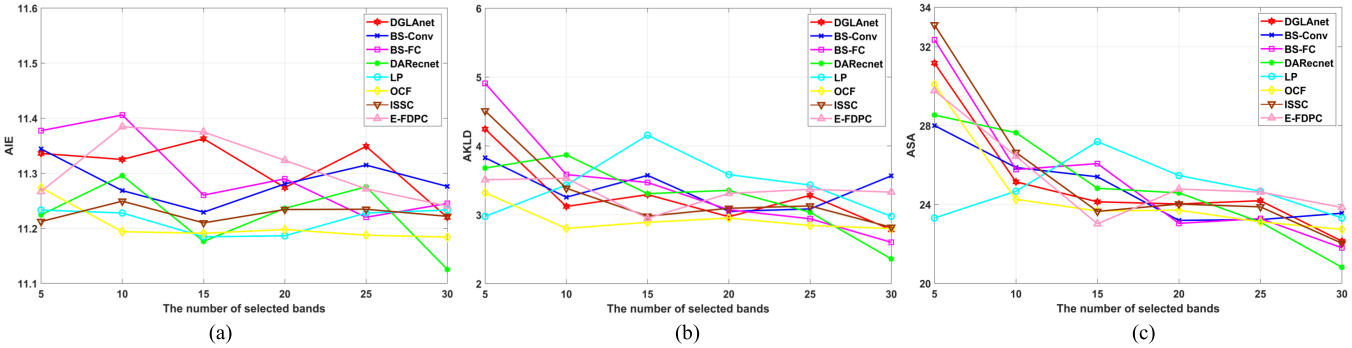


Fig. 8. Comparison in metrics of selected band subsets by different methods on the PaviaU dataset. (a) AIE. (b) AKLD. (c) ASA.

deep learning-based methods and even surpasses BS-Conv and BS-FC. In the case of the Yancheng dataset, the curves of DGLAnet, OCF, ISSC, BS-Conv, and BS-FC are close to each other. These methods are significantly better than others.

D. Quantitative Analysis of Band Subsets

In this experiment, we use average Kullback–Leibler divergence (AKLD), average spectral angle (ASA), and average information entropy (AIE) [27], [62] to make a quantitative evaluation of the bands selected by different methods, which is expressed as

$$\mathrm{AKLD} = \frac{1}{k^2}\sum_{i=1}^{k}\sum_{j=1}^{k}\mathrm{KLD}(b_i \parallel b_j) + \mathrm{KLD}(b_j \parallel b_i) \quad (18)$$

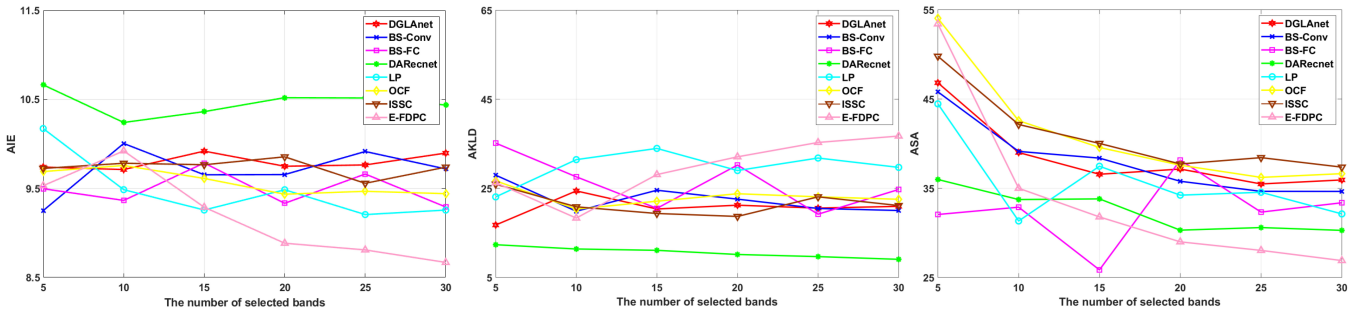
$$\mathrm{ASA} = \frac{1}{k^2}\sum_{i=1}^{k}\sum_{j=1}^{k}\mathrm{SA}(b_i, b_j) \quad (19)$$

$$\mathrm{AIE} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{IE}(b_i) \quad (20)$$

where $\mathrm{KLD}(b_i \parallel b_j) = \sum_{l=1}^{s\times s}\hat{b}_{il}\log(\hat{b}_{il}/\hat{b}_{jl})$ is the relative entropy between bands b_i and b_j, while $\hat{b}_{il}$ and $\hat{b}_{jl}$ are the corresponding normalized spectral response of the lth pixel in the patch; $\mathrm{SA}(b_i, b_j) = \arccos(b_i^T b_j/\|b_i\|\|b_j\|)$ is the spectral angle between b_i and b_j; and $\mathrm{IE}(b_i)$ is the information entropy in (14).

Figs. 7–9 show the results of eight BS methods. For the Indian Pines dataset, LP achieves the best AIE and ASA. However, its AKLD is extremely unacceptable, which leads to poor classification performance. Although the AKLD of E-FDPC is decent, its curve of OA is the lowest. The reason is that it's ASA and AIE are the smallest. The curves of DGLAnet in AIE and ASA are lower than that of LP while having the highest accuracy. The reason for this phenomenon is that DGLAnet can achieve a satisfactory result in AKLD. It can be considered that DGLAnet selects the bands with high separation while ensuring the information. For the PaviaU dataset in Fig. 8, we find that the AKLD and ASA of LP surpass those of other methods, but LP fails to find informative bands. DGLAnet has comparable values with the other methods in all the metrics. That makes DGLAnet's OA the best.

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(a) (b) (c)

Fig. 9. Comparison in metrics of selected band subsets by different methods on the Yancheng dataset. (a) AIE. (b) AKLD. (c) ASA.

TABLE V
COMPARISON OF COMPUTING TIME OF DIFFERENT METHODS

Methods	Indian Pines	PaviaU	Yancheng
E-FDPC	0.72s	4.39s	53.84s
ISSC	0.87s	0.55s	3.71s
OCF	0.65s	0.57s	6.03s
LP	15.74s	80.07s	785.73s
DARecnet	1812s	4396s	>44h
BS-Conv	601s	1713s	>18h
BS-FC	210s	1103s	>14h
DGLAnet	695s	1901s	>19h

As for Yancheng, DARecnet outperforms the others in AIE but ranks the last in both band correlation and OA. It shows that DARecnet could not guarantee low redundancy between bands. DGLAnet achieves comparable performance with ISSC and BS-Conv, dominating the others in AIE. LP is ahead of the others in AKLD, and DGLAnet obtains comparable values with the rest of the contenders. With regard to ASA, DGLAnet has comparable values with ISSC, OCF, and BS-Conv. To sum up, DGLAnet has competitive behaviors in all the metrics on the three datasets, which is conducive to the classification.

E. Analysis of Computation Complexity

Table V compares the time required by eight different methods to select 30 bands on three datasets. The computational time of deep learning-based methods is significantly longer than that of the traditional methods, but their classification accuracies outperform most traditional ones. Among four deep learning-based methods, BS-FC is the fastest of the three datasets. The computational time of DGLAnet is slightly slower than that of BS-Conv and that of DARecnet is significantly higher than the other deep learning-based methods.

The parameters in deep learning models are one of the important indicators to affect the computational complexity. As shown in Table VI, the number of parameters of BS-FC is the least. The number of parameters of DGLAnet is close to that of DARecnet. It is clear from Table VI that the parameters of the other three methods are reduced by nearly two-thirds compared to BS-Conv. More specifically, the parameters of DGLAnet, DARecnet, and BS-FC are reduced by 54.30%, 61.89%, and 73.88% on average. In spite of taking the shortest time, BS-FC is not as robust as DGLAnet in classification. The computational time of DGLAnet is only slightly higher than that of BS-Conv, but it reduces the number of parameters by more than half and has the best classification accuracy.

TABLE VI
PARAMETERS OF DEEP LEARNING-BASED METHODS

Methods	Indian Pines	PaviaU	Yancheng
DARecnet	200,976	163,934	177,810
BS-Conv	591,584	398,748	486,220
BS-FC	154,416	104,170	126,962
DGLAnet	273,772	183,733	217,350

TABLE VII
EFFECT OF DIFFERENT ATTENTION ON THE MODEL

Methods	Datasets/OA(%)		
	Indian Pines	PaviaU	Yancheng
SAnet	80.44±0.45	92.23±0.16	91.86±0.8
BAnet	80.62±0.33	91.9±0.19	91.95±1.38
GAnet	80.66±0.71	92.35±0.29	91.97±0.86
LAnet	80.56±0.51	92.47±0.17	92.06±1.11
DGLAnet	81.68±0.33	92.68±0.23	92.84±0.88

In short, DGLAnet is more practical than the other deep learning-based methods.

F. Ablation Experiments

In order to analyze the influence of different parts on the results of BS, we conduct ablation experiments. SAnet denotes that the network only uses the spatial attention module for BS, and BAnet indicates that only the spectral attention module is used. GAnet and LAnet, respectively, mean that the model only uses global and local spatial–spectral information. The number of bands selected by each method is 30, and the results are shown in Table VII.

For the Indian Pines data, DGLAnet can increase by 1.24% and 1.06% on the basis of SAnet and BAnet, respectively. This may be because the Indian Pines data contains more spatial noise, so the band information is more important. High spatial resolution and low band noise of the PaviaU dataset make SAnet superior to BAnet and even close to DGLAnet. On the Yancheng data, the improvement of the model brought by the spectral and spatial information is almost the same.

Therefore, the combined consideration of spatial and spectral information can achieve better results which indicates the importance of spatial information for BS. Besides, we compare the role of global and local information. As shown in Table VII, both global and local information have positive contributions to the model. The improvement brought by global and local information is similar. In general, it is reasonable and necessary to jointly use spatial and spectral feature extraction modules, and the global and local information needs to be integrated.

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TABLE VIII
BAND SUBSETS SELECTED BY EIGHT METHODS ON THREE DATASETS

Datasets	Methods	Band numbers
Indian Pines	DGLAnet	14, 15, 86, 85, 53, 39, 27, 127, 17, 45, 173, 64, 37, 128, 18, 35, 178, 168, 149, 67, 42, 89, 22, 79, 109
	BS-Conv	47, 34, 141, 162, 81, 36, 179, 45, 127, 37, 139, 72, 181, 67, 193, 17, 54, 153, 186, 120, 25, 29, 27, 157, 84
	DARecnet	53, 41, 75, 162, 58, 17, 127, 122, 42, 118, 7, 72, 76, 140, 16, 138, 149, 179, 100, 65, 52, 101, 125, 64, 38
	BS-FC	199, 165, 163, 160, 153, 124, 108, 107, 88, 69, 68, 62, 48, 44, 37, 33, 26, 22, 15, 9, 8, 164, 10, 71, 115
	LP	29, 42, 1, 61, 13, 75, 89, 34, 20, 57, 23, 85, 72, 26, 17, 6, 97, 68, 94, 54, 37, 117, 46, 64, 9
	OCF	8,10,11,29,40,47,49,51,54,76,83,98,107,109,112,131,133,135,150,159,167,171,184,188,200
	ISSC	69,8,77,66,49,84,86,23,44,46,68,127,138,42,22,147,151,115,162,83,47,25,179,53,100
	E-FDPC	167,124,67,152,92,50,137,179,118,125,134,142,109,105,80,107,189,106,155,111108,130,103,186,180
PaviaU	DGLAnet	97, 29, 96, 65, 85, 40, 84, 13, 101, 78, 83, 32, 71, 31, 54
	BS-Conv	91, 43, 17, 49, 72, 4, 79, 39, 81, 54, 8, 32, 5, 100, 99
	DARecnet	100, 39, 78, 99, 69, 14, 59, 32, 83, 20, 12, 22, 9, 54, 8
	BS-FC	3, 9, 97, 43, 64, 63, 38, 78, 83, 66, 1, 20, 96, 47, 87
	LP	63,95,15,1,4,74,34,83,7,86,46,102,10,18,99
	OCF	3,8,15,19,29,33,36,48,53,61,65,72,80,88,99
	ISSC	53,99,24,18,83,59,12,35,47,4,91,30,40,75,68
	E-FDPC	61,92,33,19,52,9,82,3742,90,50,56,54,48,86
Yancheng	DGLAnet	74, 30, 15, 140, 11, 77, 54, 45, 59, 133, 89, 134, 90, 37, 111, 40, 55, 39, 31, 24, 130, 91, 136, 43, 34, 35, 36, 49, 81, 50
	BS-Conv	97, 76, 19, 58, 127, 106, 56, 30, 37, 49, 86, 39, 34, 71, 98, 18, 88, 12, 93, 46, 16, 125, 21, 133, 101, 38, 20, 40, 43, 60
	DARecnet	12, 26, 68, 24, 56, 4, 36, 1, 55, 60, 89, 29, 44, 82, 64, 99, 70, 2, 27, 63, 87, 13, 39, 14, 34, 74, 16, 49, 7, 23
	BS-FC	31, 16, 69, 72, 130, 137, 35, 26, 118, 120, 17, 89, 62,119, 41, 75,67, 50, 134, 82, 105, 122, 90, 48, 111, 128, 93, 21, 125, 101
	LP	24,87,101,63,40,84,118,77,67,147,34,98,44,1,122,94,114,37,80,60,144,139,125,70,11,91,111,49,74,136
	OCF	1,4,6,11,14,20,23,28,35,38,39,40,41,46,49,57,72,75,80,94,98,99,101,104,110,115,123,130,132,137
	ISSC	95,80,42,25,19,48,41,31,144,121,4,82,32,33,112,104,22,34,51,16,137,89,92,129,56,73,64,11,28,43
	E-FDPC	129,28,47,109,6,92,94,57,72,115,82,104,138,141,66,145,143,136,121,54,139,147,48,99,43,144,122,124,126,119

IV. CONCLUSION

This article proposes an end-to-end UBS-Net called DGLAnet. It uses bi-directional relations to characterize global features. It not only relaxes the limitation of the convolution kernel to obtain global information, but also reduces the number of parameters. In addition, the DGLAnet employs the max-pooling layer and the mean-pooling layer to extract texture features and background features to obtain local features. Then global and local features are used to obtain attention weights. Finally, the DGLAnet utilizes the constructed band evaluation index to select the best band from the reweighted HSI. Experimental results demonstrate that it can attain better performances than the comparison methods. When the number of bands reaches a certain number, its result is better than using all bands. Moreover, with no significant rise in the computational time, the DGLAnet has fewer parameters, which makes it more practical. The results of ablation experiments show that both spatial–spectral and global–local information are crucial in BS. Table VIII lists the bands selected by different methods for Indian Pines, PaviaU, and Yancheng datasets. In the future, we will focus on further improving the classification performance of our DGLAnet with more advanced schemes, for example, investigating active learning and semisupervised models, combining with spatial–spectral classifiers, and the lightweight deep learning networks will also be explored to enhance the computational efficiency.

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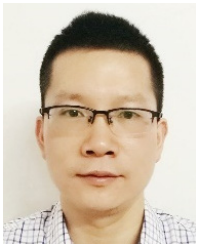
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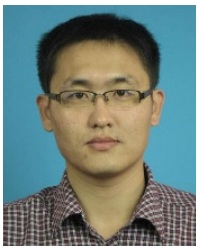
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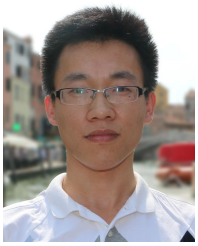
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